

December 2024

(For AI&ML)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

1. Which of the following is NOT a key component of machine learning?
A) Data
B) Algorithms
C) Hardware
D) User interfaces
2. Which of the following are not ML methods?
A) Supervised Learning
B) Reinforcement Learning
C) Semi Machine Learning
D) Unsupervised Learning
3. Which machine learning task is focused on finding patterns and relationships in data?
A) Classification
B) Regression
C) Unsupervised Learning
D) Reinforcement Learning
4. How do you handle missing or corrupted data in a dataset?
A) Drop missing rows or columns
B) Replace missing values with mean/median/mode
C) Assign a unique category to missing values
D) All of the above
5. ML is a field of AI consisting of learning algorithms that
A) Improve their performance
B) At executing some task
C) Over time with experience
D) All of the above
6. What is the difference between supervised and unsupervised learning?
A) Supervised learning requires labeled data while unsupervised learning does not.
B) Unsupervised learning requires labeled data while supervised learning does not.
C) Supervised learning does not require data while unsupervised learning does.
D) There is no difference between supervised and unsupervised learning.
7. Which of the following is NOT a type of data attribute?
A) Numeric
B) Categorical
C) Boolean
D) Time Series
8. What is the purpose of feature scaling in machine learning?
A) To convert categorical features into numerical features
B) To reduce the dimensionality of the feature space
C) To standardize the range of numerical features
D) To introduce new features into the model
9. Candidate generation and pruning in Apriori are used to
A) Find infrequent itemsets
B) Reduce the size of the itemset space
C) Calculate confidence values
D) Identify outliers in the data
10. What is the key assumption of the Naïve Bayes classifier?
A) Features are dependent on each other
B) Features are independent of each other
C) Features are continuous
D) Features are normally distributed
11. What will happen if support is reduced?
A) Number of frequent item sets remains the same
B) Some item sets will add to the current set of frequent item sets
C) Some item sets will become infrequent while others will become frequent
D) Can not say

12. Which of the following is finally produced by Hierarchical Clustering?
 A) final estimate of cluster centroids B) tree showing how close things are to each other
 C) assignment of each point to clusters D) all of the mentioned
13. Which of the following is required by K-means clustering
 A) defined distance metric B) number of clusters
 C) initial guess as to cluster centroids D) all of the mentioned
14. Simple linear regression is used to model the relationship between:
 A) Two or more independent variables and a dependent variable B) A single independent variable and a dependent variable
 C) Multiple dependent variables and one independent variable D) Categorical variables and a continuous variable
15. What is the primary goal of a Support Vector Machine (SVM)?
 A) Maximize the margin between data points B) Minimize the margin between data points
 C) Fit a line through all data points D) Maximize the number of support vectors
16. Which of the following is an example of a clustering algorithm?
 A) Decision tree B) Random forest
 C) K-means D) Gradient descent
17. Which of the following is a common approach to reducing dimensionality in machine learning?
 A) Feature selection B) Feature extraction
 C) Feature scaling D) Feature engineering
18. Which unsupervised learning algorithm is used for dimensionality reduction?
 A) K-Means clustering B) Principal Component Analysis
 C) Decision Tree D) K-Nearest Neighbors.
19. When performing regression or classification, which of the following is the correct way to preprocess the data?
 A) Normalize the data -> PCA -> training B) PCA -> normalize PCA output -> training
 C) Normalize the data -> PCA -> normalize PCA output -> training D) None of the above
20. Which of the following is a widely used and effective machine learning algorithm based on the idea of bagging?
 A) Decision Tree B) Regression
 C) Classification D) Random Forest

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) What is machine Learning. Explain different types of machine learning.
 b) List the different types of attributes based on measurement scale.
 c) Illustrate the different types of ordered data with an example.
2. a) Write Hunts algorithm and apply the same to construct decision tree for the following data.

Marks	BT*	CO*	PO*
8	L*2	1	1
4	L2	1	1
4	L2	1	1

	binary	categorical	continuous	class
Tid	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- b) Consider the following values in confusion matrix and calculate accuracy and Error rate.

$$TP = 45 \quad TN = 23 \quad FP = 15 \quad FN = 32$$

8	L2	2	1
8	L2	2	1

3. a) Compute Gini index and entropy for the following Nominal attribute (CarType). Identify the best split.

Car Type

{Sports, Luxury}
{Family}

Car Type

{Family, Luxury}
{Sports}

Car Type

Family
Sports
Luxury

Car Type

{Sports, Luxury}
{Family}

Car Type

{Sports}
{Family, Luxury}

Car Type

Family
Sports
Luxury

C0

9

1

C1

7

3

C0

8

2

C1

0

10

C0

1

8

1

C1

3

0

7

8 L2 2 1

- b) Write the algorithm for KNN and predict the following test data based on KNN.

Test Data (T): <Acid Durability=4, Strength=8, Class=?>

Name	Acid Durability	Strength	Class
Type 1	4	3	Good
Type 2	6	5	Bad
Type 3	3	7	Bad
Type 4	2	8	Good

8 L3 2 2

Unit - II

4. a) Explain the process of Frequent Itemset Generation using the Apriori Algorithm.
- b) For the following dataset, compute frequent itemsets with minimum support of 40% using an Apriori algorithm.

8 L3 3 2

TID	Items
1	{a, d, e}
2	{a, b, c, e}
3	{a, b, d, e}
4	{a, c, d, e}
5	{b, c, e}
6	{b, d, e}
7	{c, d}
8	{a, b, c}
9	{a, d, e}
10	{a, b, e}

8 L3 3 2

5. a) Write the Importance of Support vector machine in machine learning.
- b) Explain the different types of clusters

8 L2 3 1

8 L2 3 1

6. a) Illustrate DBSCAN algorithm with an example
- b) For the given proximity matrix, construct dendrogram and nested cluster representation using single link (MIN) agglomerative hierarchical clustering (AHC) method

8 L2 4 1

Dist	A	B	C	D	E	F
A	0.00	0.71	5.66	3.61	4.24	3.20
B	0.71	0.00	4.95	2.92	3.54	2.50
C	5.66	4.95	0.00	2.24	1.41	2.50
D	3.61	2.92	2.24	0.00	1.00	0.50
E	4.24	3.54	1.41	1.00	0.00	1.12
F	3.20	2.50	2.50	0.50	1.12	0.00

8 L2 4 2

Unit – III

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|----|----|---|---|----|---|---|
| 7. | a) | With an application of random forest, explain working of random forest in detail. | 8 | L2 | 5 | 1 |
| | b) | Write the difference between bagging and boosting. | 8 | L2 | 5 | 1 |
| 8. | a) | Write how to use factor analysis to reduce the dimensions. | 8 | L2 | 5 | 1 |
| | b) | Explain different method of solving class imbalance problem. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
